Distance formula

It is the formula used for finding the distance between 2 points.

Mathematically:-

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Derivation

It is derived using the pythagorus theorem. In rt-angled $\triangle ABC$, we have

By Pythagoras's theorem,

$$AB^2 = AC^2 + BC^2$$

Distance between point A and C is $(x_2 - x_1)$

Distance b/w point B and C is $(y_2 - y_1)$

$\therefore$ Distance is calculated as

$$d^2 = (x_2 - x_1)^2 + (y_2 - y_1)^2$$

$$\boxed{d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}}$$

In 3D Distance formula is

$$\boxed{d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}}$$

Distance b/w two points in polar Coordinates

Lets take two points $A(r_1, \theta_1)$ and $B(r_2, \theta_2)$ then the distance b/w them is calculated using the Distance formula.

$$AB = \sqrt{(r_1)^2 + (r_2)^2 - 2r_1r_2\cos(\theta_1 - \theta_2)}$$

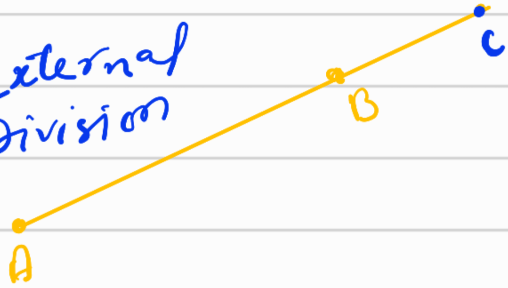
SECTION FORMULA

It is used to find the coordinates of a point that divides the line in a given ratio or to find the ratio in which the line is divided by a point of given coordinates.

Any point can divide a line segment in 2 ways, either the point can be on the line segment and divide the line internally or the point can be on the extended line segment and divide the line segment externaly.

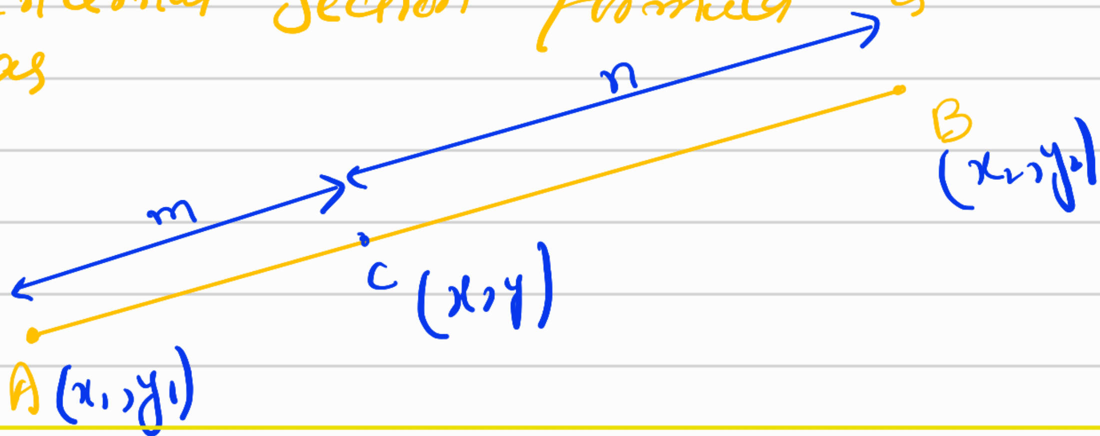
Internal division

External Division



When the point divides the line segment in the ratio $m:n$ internally at point C then that point lies on the line segment i.e C divides AB internally.

Then the Internal Section formula is used. If the coordinates of A and B are (x_1, y_1) and (x_2, y_2) respectively then Internal Section Formula is given as



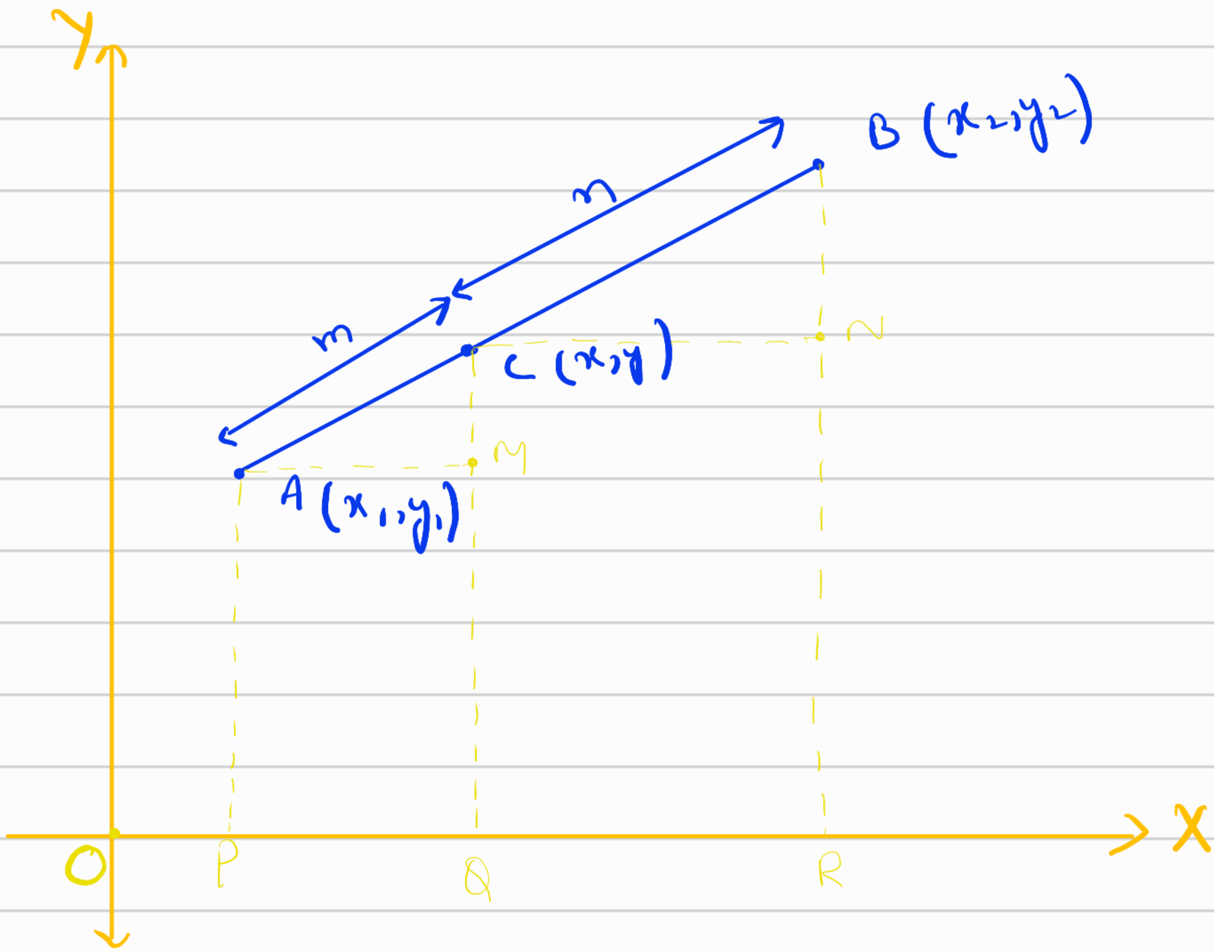
$$\boxed{C(x,y) = \left\{ \frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n} \right\}}$$

Derivation of Section formula.

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the end points of the given line segment AB and $C(x, y)$ be the point that internally divides AB in the ratio $m : n$.

i.e. $AC/CB = m/n$

We want to find the coordinates of $C(x, y)$. Take the given line segment and draw perpendiculars of A, C and B parallel to Y coordinate joining at P, Q and R on the X-axis as:



From above diagram

$AM = PQ = OQ - OP = (x - x_1)$

$CN = QR = OR - OQ = (x_2 - x)$

$CM = CQ - MQ = (y - y_1)$

$BN = BR - NR = (y_2 - y)$

Clearly, $\triangle AMC \sim \triangle CNB$ (as angles are equal for both $\triangle$'s)

$\therefore$, ratio of sides are same for similar $\triangle$'s

$\Rightarrow AC/CB = AM/CN = CM/BN$

$\Rightarrow m/n = (x - x_1)/(x_2 - x) = (y - y_1)/(y_2 - y)$

$m/n = (x - x_1)/(x_2 - x)$ and $m/n = (y - y_1)/(y_2 - y)$

Solving Ist condition.

$m(x_2 - x) = n(x - x_1)$

$\Rightarrow mx_2 - mx = nx - nx_1$

$\Rightarrow mx_2 + nx_1 = mx + nx$

$\Rightarrow mx_2 + nx_1 = x(m+n)$

$$\Rightarrow \boxed{x = \frac{mx_2 + nx_1}{m+n}}$$

Solving 2nd condition

$$m(y_2 - y) = n(y - y_1)$$

$$\Rightarrow my_2 - my = ny - ny_1$$

$$\Rightarrow my_2 + ny_1 = ny + my$$

$$\Rightarrow my_2 + ny_1 = y(n+m)$$

$$\Rightarrow \boxed{y = \frac{my_2 + ny_1}{n+m}}$$

$$\therefore \boxed{C(x, y) = \frac{mx_2 + nx_1}{m+n}, \frac{my_2 + my_1}{m+n}}$$

External Section Formula

When the point which divides the line segment in the ratio $m:n$ lies outside the line segment i.e, when we extend the line it coincides with the point then we use the external Division. If the coordinates of A and B are (x_1, y_1) and (x_2, y_2) respectively then external section formula is used. It is given as

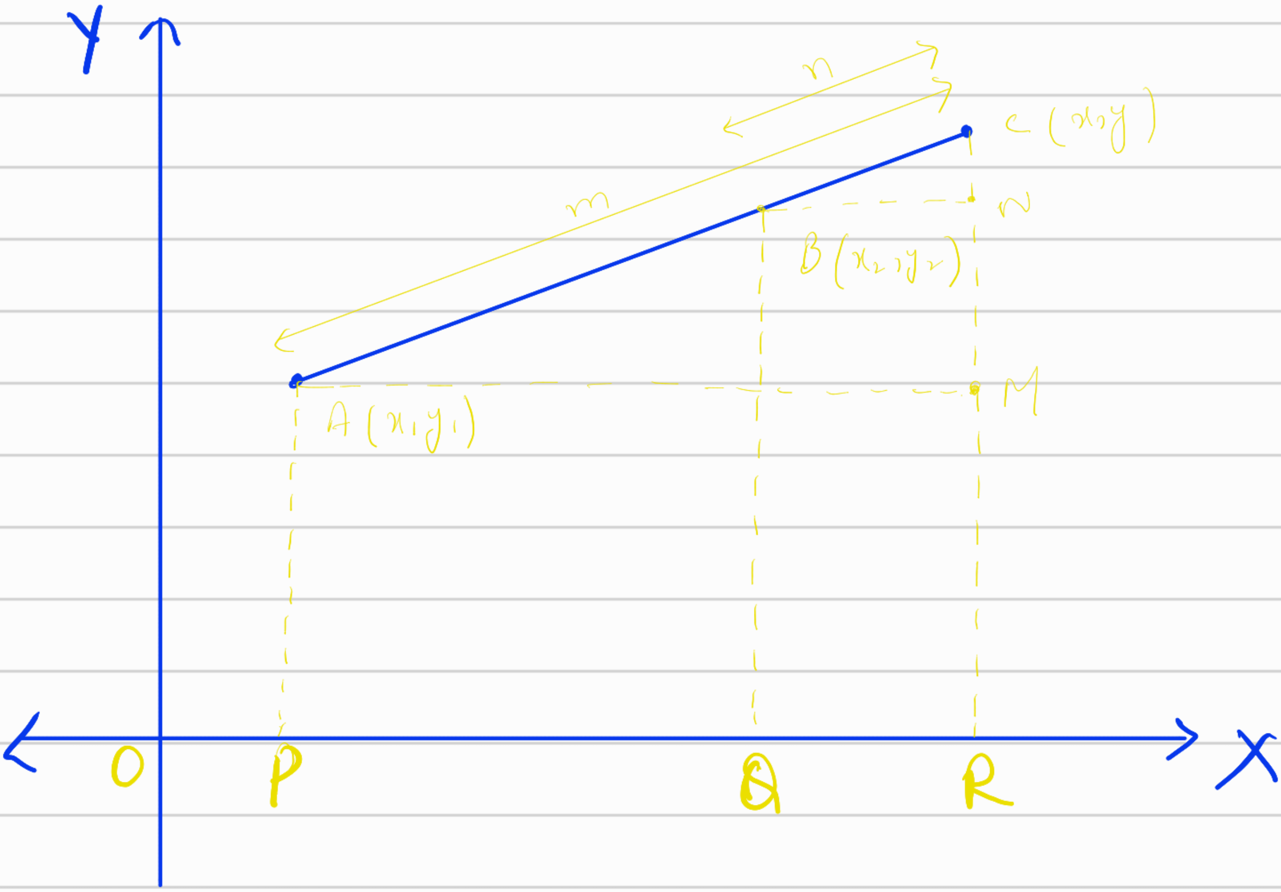
$$C(x,y) = \left\{ \frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right\}$$

Derivation

Here we have to take point $C(x,y)$ outside the line segment.

Let $A(x_1, y_1)$ and $B(x_2, y_2)$ be the end points of the given line segment AB and $C(x,y)$ be the point that divides AB in the ratio $m:n$ externally.

To find the coordinates (x,y) of C take the external divided line AB and draw $\perp$'s from A, B and C $\parallel$ to the y-axis joining at P, Q and R on the x-axis and $\perp$ from A and B parallel to x-axis which meets the $\perp$ draw from C $\parallel$ to y-axis at M and N.



from the diagram

$AM = PR = OR - OP = (x - x_1)$

$BN = QR = OR - OQ = (x - x_2)$

$CM = RC - MR = (y - y_1)$

$CN = CR - NR = (y - y_2)$

Clearly, $\triangle AMC \sim \triangle CNB$ (as all angles are equal for both $\triangle$'s)

as, the ratio of all the sides are same for similar $\triangle$'s

$AC/BC = AM/BN = CM/CN$

$m/n = (x - x_1)/(x - x_2) = (y - y_1)/(y - y_2)$

$\Rightarrow m/n = (x-x_1)/(x-x_2)$ and $m/n = (y-y_1)/(y-y_2)$

Solving Ist condition.

$$m(x-x_2) = n(x-x_1)$$

$$\Rightarrow mx - mx_2 = nx - nx_1$$

$$\Rightarrow mx - nx = mx_2 - nx_1$$

$$\Rightarrow x(m-n) = mx_2 - nx_1$$

$$\Rightarrow \boxed{x = \frac{mx_2 - nx_1}{m-n}}$$

Solving 2nd condition.

$$m(y-y_2) = n(y-y_1)$$

$$\Rightarrow my - my_2 = ny - ny_1$$

$$\Rightarrow my - ny = my_2 - ny_1$$

$$\Rightarrow y(m-n) = my_2 - ny_1$$

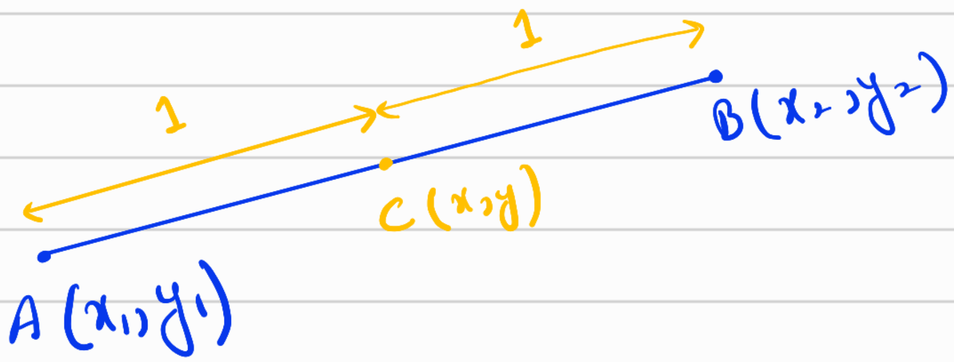
$$\Rightarrow \boxed{y = \frac{my_2 - ny_1}{m-n}}$$

$$\therefore C(x,y) = \left\{ \frac{mx_2 - nx_1}{m-n}, \frac{my_2 - ny_1}{m-n} \right\}$$

Section formula for Mid point

when the pt dividing the line segment joining the two pts. A and B coincide with the mid point of the line segment. then the special case emerges. The section formula for this case is also reffered to as the midpoint formula.

Let 2 points $A(x_1, y_1)$ and $B(x_2, y_2)$ and a point $C(x, y)$ be the mid point of the line segment AB i.e, $AC : CB = 1 : 1$,



then the section formula becomes.

$$\boxed{C(x, y) = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)}$$